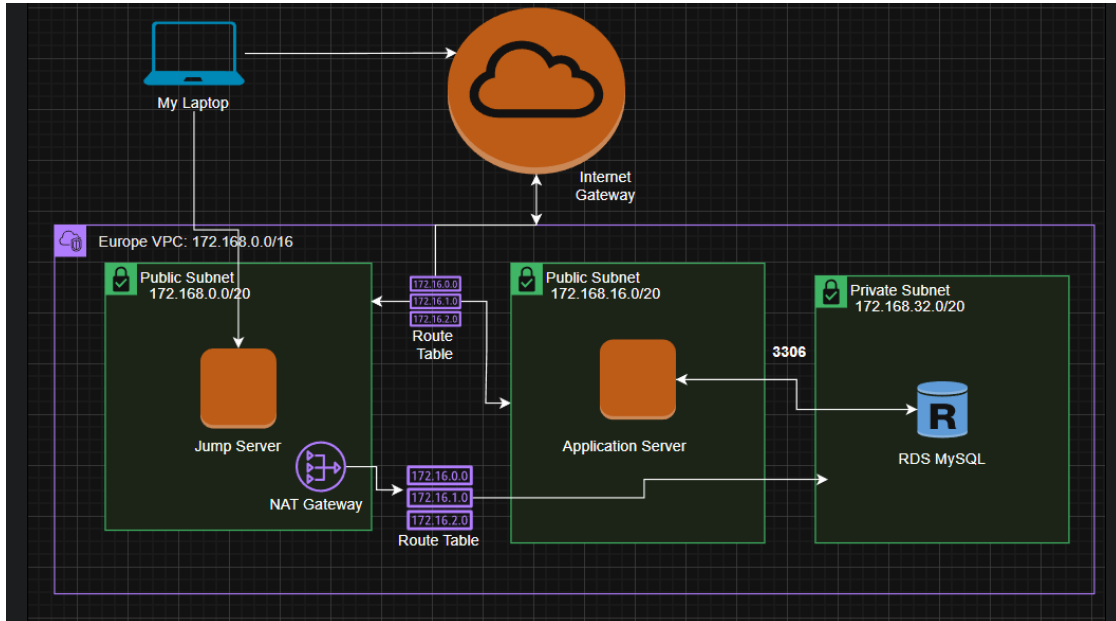


2-Tier Architecture

Title: 2-tier Architecture - java app on tomcat --RDS database in backend tomcat installation.



1. Create VPC

AWS Console → **VPC** → “Create VPC”.

Name: studentapp-vpc

CIDR: 172.168.0.0/16

Tenancy: default → Create.

The screenshot shows the 'Create VPC' page in the AWS Management Console. The 'VPC settings' section is active, showing 'Resources to create' with 'VPC only' selected. The 'Name tag - optional' field contains 'studentapp-vpc'. Under 'IPv4 CIDR block', 'IPv4 CIDR manual input' is selected, and the 'IPv4 CIDR' field contains '172.168.0.0/16'. The 'IPv6 CIDR block' section shows 'No IPv6 CIDR block' selected.

2. Create Subnets

Create three subnets in this VPC (you can keep all in same AZ for lab):

- **Public Subnet 1 – Jump**

- Name: Public-Jump
- CIDR: 172.168.0.0/20

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Subnets > Create subnet'. The page title is 'Subnet 1 of 1'. Under 'Subnet name', there is a text input field containing 'Public-Jump'. Under 'Availability Zone', a dropdown menu is set to 'No preference'. Under 'IPv4 VPC CIDR block', a dropdown menu is set to '172.168.0.0/16'. Under 'IPv4 subnet CIDR block', a text input field contains '172.168.0.0/20', and a tooltip indicates '4,096 IPs'. At the bottom, there is a section for 'Tags - optional'.

The screenshot shows the 'Edit subnet settings' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Subnets > subnet-003e9fd593c7a86d9 > Edit subnet settings'. The page title is 'Edit subnet settings'. Under the 'Subnet' section, the 'Subnet ID' is 'subnet-003e9fd593c7a86d9' and the 'Name' is 'Public-Jump'. Under the 'Auto-assign IP settings' section, the 'Enable auto-assign public IPv4 address' checkbox is checked. Under the 'Resource-based name (RBN) settings' section, there is a description: 'Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.'

- **Public Subnet 2 – App**

- Name: Public-App
- CIDR: 172.168.16.0/20

VPC > Subnets > Create subnet

Subnet 2 of 3

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public-App
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.168.0.0/16

IPv4 subnet CIDR block
172.168.16.0/20 4,096 IPs

VPC > Subnets > subnet-02287435896568df7 > Edit subnet settings

Edit subnet settings [Info](#)

Subnet

Subnet ID

subnet-02287435896568df7

Name

Public-App

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)
Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

• Private Subnet – DB

- Name: Private-Db
- CIDR: 172.168.32.0/20

Account ID: 0950-7410-7161

Sakshi Jain

VPC > Subnets > Create subnet

Subnet 3 of 3

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private-Db
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.168.0.0/16

IPv4 subnet CIDR block
172.168.32.0/20 4,096 IPs

▼ Tags - optional

Key

Value - optional

https://eu-north-1.console.aws.amazon.com/console/home?region=eu-nort...

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[Privacy](#)

[Terms](#)

[Cookie preferences](#)

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private-Db-2
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Europe (Stockholm) / eu-north-1a

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.168.0.0/16

IPv4 subnet CIDR block
172.168.48.0/20 4,096 IPs

3. Internet Gateway

1. VPC → **Internet Gateways** → Create.
2. Name: studentapp-igw
3. Attach it to studentapp-vpc.

VPC > Internet gateways > Create internet gateway

Create internet gateway

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.
studentapp-igw

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	studentapp-igw	Remove

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

VPC > Internet gateways > Attach to VPC (igw-0b4301b550f120a76)

✓ The following internet gateway was created: igw-0b4301b550f120a76 - studentapp-igw. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#)

Attach to VPC (igw-0b4301b550f120a76)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.
vpc-0bf8e99626618ff53

AWS Command Line Interface command

[Cancel](#) [Attach internet gateway](#)

4. Route Tables

Public route table (for both public subnets)

1. VPC → Route tables → Create.
2. Name: public-rt
3. Associate with VPC studentapp-vpc.
4. Edit routes:
 - Add route: 0.0.0.0/0 → Internet Gateway (studentapp-igw).
5. Subnet associations:
 - Attach **public-jump** and **public-app** to public-rt.

The screenshot shows two panels from the AWS Management Console. The top panel, 'Edit routes', displays a table of routes for route table 'rtb-00c57b5f731ba8705'. It shows a route for destination '172.168.0.0/16' with target 'local' and status 'Active'. Below this, there are search boxes and dropdowns to add a new route for '0.0.0.0/0' with target 'Internet Gateway' (igw-0b4301b550f120a76). The bottom panel, 'Edit subnet associations', shows a table of available subnets. It lists 'Public-Jump' (subnet-003e9fd593c7a86d9) and 'Public-App' (subnet-02287435896568df7) as selected subnets. The 'Private-Db' subnet is not selected.

Destination	Target	Status	Propagated	Route Origin
172.168.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public-Jump	subnet-003e9fd593c7a86d9	172.168.0.0/20	-	Main (rtb-00c57b5f731ba8705 / p
Private-Db	subnet-0a486c4169dcb4b2f	172.168.32.0/20	-	Main (rtb-00c57b5f731ba8705 / p
Public-App	subnet-02287435896568df7	172.168.16.0/20	-	Main (rtb-00c57b5f731ba8705 / p

Create NAT Gateway

The screenshot shows the 'Create NAT gateway' page in the AWS Management Console. It includes sections for 'NAT gateway settings', 'Availability mode', 'Subnet', and 'Connectivity type'. The 'Name' is set to 'public-jump-subnet'. The 'Availability mode' is set to 'Zonal'. The 'Subnet' is set to 'subnet-0ec496f2d2476f1bd (Public-Jump)'. The 'Connectivity type' is set to 'Public'.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
public-jump-subnet

Availability mode
Choose whether to deploy across all zones in the region or restrict to a single availability zone.
☐ Regional - new
☒ Zonal

Subnet
Select a subnet in which to create the NAT gateway.
subnet-0ec496f2d2476f1bd (Public-Jump)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

[Allocate Elastic IP](#)

► **Additional settings** [Info](#)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key [×](#) **Value - optional** [×](#) [Remove](#)

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create NAT gateway](#)

Private route table (for DB subnet)

For simple lab you can skip NAT and leave private subnet with no Internet, or add NAT if you like. I'll include it:

1. Create route table private-rt.
2. Associate with VPC studentapp-vpc.
3. Subnet associations: attach private-db-subnet.
4. We will point 0.0.0.0/0 → NAT-Gateway

☰ [VPC](#) > [Route tables](#) > Create route table 🔍 ⚙️

Create route table [Info](#)
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key [×](#) **Value - optional** [×](#) [Remove](#)

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create route table](#)

☰ [VPC](#) > [Route tables](#) > [rtb-0f7e5e75c12877b2f](#) > Edit subnet associations 🔍 ⚙️

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (2/4)

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Private-Db	subnet-066f033a30d191d98	172.168.32.0/20	-	rtb-0f7e5e75c12877b2f / private-rt
☐	Public-Jump	subnet-0ec496f2d2476f1bd	172.168.0.0/20	-	rtb-00c854cedc6725b4b / public-rt
☐	Public-App	subnet-0f4efb34aca1bfa89	172.168.16.0/20	-	rtb-00c854cedc6725b4b / public-rt
☒	Private-Db-2	subnet-0b595e54d730a49cb	172.168.48.0/20	-	Main (rtb-00c854cedc6725b4b / public...

Selected subnets

[Cancel](#) [Save associations](#)

VPC > Route tables > rtb-0f7e5e75c12877b2f > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
172.168.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0 X	NAT Gateway	-	No	CreateRoute
	Q nat-07e20eb6bd185c94 X			

Add route

Cancel Preview Save changes

❖ Security Groups

Go to EC2 → Security Groups.

5. SG for Jump Server – sg_jump

- VPC: studentapp-vpc
- Inbound:
 - SSH (22) -0.0.0.0/0
- Outbound: allow all (default).

EC2 > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name info

sg_jump

Name cannot be edited after creation.

Description info

Jump Server

VPC info

vpc-04927db60e5a76e81 (studentapp-vpc)

Inbound rules

Security group rule ID

sgr-0bc8100967b7ef2e3

Type info

SSH

Protocol info

TCP

Port range info

22

Source info

Anywh...

Q 0.0.0.0/0

0.0.0.0/0 X

Description - optional info

Delete

Add rule

You have not made any changes.

Cancel Preview changes Save rules

6. SG for Application Server – sg_app

- Inbound:
 - HTTP / Custom TCP 8080 from 0.0.0.0/0 (so browser can hit Tomcat)
 - SSH (22) from sg-jump (not from internet)
- Outbound:
 - All (default).

The screenshot shows the 'Create security group' page in the AWS Management Console. The 'Basic details' section includes the security group name 'sg_app', description 'app server', and VPC 'vpc-04927db60e5a76e81 (studentapp-vpc)'. The 'Inbound rules' section shows two rules: one for port 8080 and another for port 22, both allowing traffic from '0.0.0.0/0' on TCP. Buttons for 'Add rule', 'Cancel', 'Preview changes', and 'Save rules' are visible.

7. SG for RDS – sg_rds

- Inbound:
 - MySQL/Aurora (3306) from sg-app only.
- Outbound:
 - All (default).

Now only Tomcat EC2 can talk to the database.

The screenshot shows the 'Create security group' page for 'sg_rds'. The 'Basic details' section has the name 'sg_rds', description 'rds', and VPC 'vpc-04927db60e5a76e81 (studentapp-vpc)'. The 'Inbound rules' section shows a rule for MySQL/Aurora on port 3306, allowing traffic from 'sg-08fcd4b1fcd8d85' and 'sg-08fcd4b1fcd8d893d'. Buttons for 'Add rule', 'Cancel', 'Preview changes', and 'Save rules' are visible.

Create Key Pair

1. EC2 → Key pairs → Create key pair.
2. Name: studentapp-key
3. Type: RSA, .pem (for Linux).
4. Download vpc.pem to your laptop.

8. Launch Jump Server EC2

9. Launch Application Server EC2

10. Create DB Subnet Group

Aurora and RDS > **Subnet groups** > Create DB subnet group

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

eu-north-1a X eu-north-1b X

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

Private-Db-2 Subnet ID: subnet-0b595e54d730a49cb CIDR: 172.168.48.0/20 Private-Db Subnet ID: subnet-066f033a30d191d98 CIDR: 172.168.32.0/20

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
eu-north-1a	Private-Db-2	subnet-0b595e54d730a49cb	172.168.48.0/20

11. Create RDS MySQL in Private Subnet

Create RDS Instance:

Aurora and RDS > **Databases** > Create database

Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) ☐ Aurora (PostgreSQL Compatible) ☐ MySQL ☐ PostgreSQL

☒ MariaDB ☐ Oracle ☐ Microsoft SQL Server ☐ IBM Db2

Engine version [Info](#)

Aurora and RDS > **Databases** > Create database

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength **Strong**

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Aurora and RDS > **Databases** > Create database

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

studentapp-vpc (vpc-04927db60e5a76e81)
4 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

Aurora and RDS > Databases > Create database

DB subnet group info
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

studentapp-db-subnet-group
2 Subnets, 2 Availability Zones

Public access info
☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Existing VPC security groups
Choose one or more options

sg_rds

Goto VPC Security Group

Aurora and RDS > Databases > database-1

Aurora and RDS
Dashboard
Databases
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies

Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations

database-1
CPU
Class
db.t4g.micro
Instance
Creating
Current activity
MariaDB
Region & AZ
eu-north-1a

Connectivity & security | Monitoring | Logs & events | Configuration | Maintenance & backups | Data migrations

Connectivity & security

Endpoint & port
Endpoint
Port

Networking
Availability Zone
eu-north-1a
VPC
vpc-09b6b8ffa12c928
Subnet group
default-vpc-09b6b8ffa12c928
Subnets
subnet-04712f4123f4a8913
eu-north-1a-subnet-04712f4123f4a8913

Security
VPC security groups
studentapp-db-subnet-group (sg-0883ad1294c620e54)
Active
Publicly accessible
No
Certificate authority
rds-ca-rsa2048-g1
Certificate authority data

Edit inbound rule

EC2 > Security Groups > sg-0883ad1294c620e54 - studentapp-db-subnet-group > Edit inbound rules

Edit inbound rules info
Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules info

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-0e9cd73c0a9a1735	MYSQL/Aurora	TCP	3306	Any...	0.0.0.0/0

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel **Preview changes** **Save rules**

12. SSH & Key Sharing via Jump Server

On Command Prompt

scp -i vpc.pem vpc.pem ec2-user@13.53.200.62:/home/ec2-user Use Jump Server IP

```
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sjaian403>cd downloads/

C:\Users\sjaian403\Downloads>dir *.pem
Volume in drive C is OSDisk
Volume Serial Number is B29F-333D

Directory of C:\Users\sjaian403\Downloads

12-Oct-25 10:31 AM          1,678 download.pem
11-Nov-25 10:53 PM          1,678 Practice.pem
26-Nov-25 03:29 PM          1,674 vpc.pem
               3 File(s)      5,030 bytes
               0 Dir(s)  13,130,452,992 bytes free

C:\Users\sjaian403\Downloads>scp -i vpc.pem vpc.pem ec2-user@13.53.200.62:/home/ec2-user
The authenticity of host '13.53.200.62 (13.53.200.62)' can't be established.
ED25519 key fingerprint is SHA256:dPJT3EPjFRX822U7gvPp/IVLZmzFmLM1bnGicUaSI2K.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '13.53.200.62' (ED25519) to the list of known hosts.
vpc.pem
100% 1674 7.3KB/s 00:00

C:\Users\sjaian403\Downloads>
```

EC2 > Instances > i-02f4dea4af06aaf14 > Connect to instance

Connect

Info

Connect to an instance using the browser-based client.

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-02f4dea4af06aaf14 (jump-server)

Connection type

☒ Connect using a Public IP

☐ Connect using a Private IP

Connect using a public IPv4 or IPv6 address

Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address

☐ Public IPv6 address

☐ Private IPv4 address

☐ Private IPv6 address

13.53.200.62

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

```
chmod 400 vpc.pem
```

```
[ec2-user@ip-172-168-13-180 ~]$ ls
vpc.pem
[ec2-user@ip-172-168-13-180 ~]$ chmod 400 vpc.pem
[ec2-user@ip-172-168-13-180 ~]$
```

```
ssh -i vpc.pem ec2-user@<App_Server_Public_IP>
```

app-server

i-08322da0ae106f62

Running

t3.micro

5/3 checks passed

View alarms +

eu-north-1b

i-083222da0ae106f62 (app-server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary Info

Instance ID
i-083222da0ae106f62

Public IPv4 address

51.20.31.187 open address ↗

Private IPv4 addresses

172.168.25.106

```
[ec2-user@ip-172-168-13-180 ~]$ ls
vpc.pem
[ec2-user@ip-172-168-13-180 ~]$ ssh -i vpc.pem ec2-user@51.20.31.187
ssh: connect to host 51.20.31.187 port 22: Connection timed out
[ec2-user@ip-172-168-13-180 ~]$ ssh -i vpc.pem ec2-user@51.20.31.187
The authenticity of host '51.20.31.187 (51.20.31.187)' can't be established.
ED25519 key fingerprint is SHA256:eB7kKuCyluzvipnzlrWGPCw6fYNav7pLuRB6lZ0aMfk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '51.20.31.187' (ED25519) to the list of known hosts.

      _   _          _   _
     / \   \        / \   \       _   _
    /___\   ___    /___\   \__ _/ \_/_
   |   |  / __|  /   |   \|   | / __|
   |___| \___| \___|___|___|\___|\___|
                                     _   _
                                    / \   \
                                   /___\   \__ _/ \_/_
                                  |   |  / __|  /   |   \|   | / __|
                                  |___| \___| \___|___|___|\___|\___|

           Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-168-25-106 ~]$
```

13. Create Database & Table in RDS

EC2 > Instances > i-083222da0ae106f62 > Connect to instance

Connect to instance using the browser-based client.

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

+083222da0ae106f62

app-server

Connection type

☒ Connect using a Public IP

☐ Connect using a Private IP

Connect using a public IPv4 or IPv6 address

Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address

☐ IPv6 address

51.20.31.187

—

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

yum install mariadb105* -y

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-168-26-33 ~]$ sudo -i
[root@ip-172-168-26-33 ~]# yum install mariadb105* -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
284 kB/s | 29 kB    00:00

Package Architecture Version Repository Size
-----
Installing:
mariadb105 x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 1.5 M
mariadb105-backup x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 6.0 M
mariadb105-connect-engine x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 517 M
mariadb105-cracklib-password-check x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 13 M
mariadb105-devel x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 1.0 M
mariadb105-errmsg x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 212 M
mariadb105-gssapi-server x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 15 M
mariadb105-ogggraph-engine x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 75 M
mariadb105-pam x86_64 3:10.5.29-1.amzn2023.0.1 amazonlinux 21 M
```

Copy endpoint of RDS

The screenshot displays the AWS Management Console for an Amazon RDS instance named 'database-1'. The 'Connectivity & security' tab is selected, showing the endpoint 'database-1.c9wsou0ei4j.eu-north-1.rds.amazonaws.com' and port '3306'. A tooltip indicates the endpoint has been copied. The 'Networking' section shows the VPC 'vpc-09b6b8ffa12c928' and Subnet 'subnet-eu-north-1a7176417746a8012'. The 'Security' section shows the VPC security group 'studentapp-db-subnet-group (sg-0883ad7294c620e54)' is active and publicly accessible.

mysql -h (RDS database endpoint) -u (RDS database user) -p(RDS database user password)

```
[root@ip-172-168-25-106 ~]# mysql -h database-1.c9wsou0ei4j.eu-north-1.rds.amazonaws.com -u admin -psakshii23
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 157
Server version: 11.4.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

show databases;

create database studentapp;

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| innodb |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.001 sec)

MariaDB [(none)]> create database studentapp;
Query OK, 1 row affected (0.005 sec)

MariaDB [(none)]>
```

use studentapp;

```
CREATE TABLE if not exists students(student_id INT NOT NULL AUTO_INCREMENT,

    student_name VARCHAR(100) NOT NULL,

    student_addr VARCHAR(100) NOT NULL,

    student_age VARCHAR(3) NOT NULL,

    student_qual VARCHAR(20) NOT NULL,

    student_percent VARCHAR(10) NOT NULL,

    student_year_passed VARCHAR(10) NOT NULL,

    PRIMARY KEY (student_id)

);
```

```
MariaDB [(none)]> use studentapp;
Database changed
MariaDB [studentapp]> CREATE TABLE if not exists students(student_id INT NOT NULL AUTO_INCREMENT,
-> student_name VARCHAR(100) NOT NULL,
-> student_addr VARCHAR(100) NOT NULL,
-> student_age VARCHAR(3) NOT NULL,
-> student_qual VARCHAR(20) NOT NULL,
-> student_percent VARCHAR(10) NOT NULL,
-> student_year_passed VARCHAR(10) NOT NULL,
-> PRIMARY KEY (student_id)
-> );
Query OK, 0 rows affected (0.020 sec)

MariaDB [studentapp]> select * from students;
Empty set (0.001 sec)

MariaDB [studentapp]>
```

```
MariaDB [studentapp]> exit
Bye
```

14. Install Tomcat & App on Application Server

Run these commands on app-server EC2

yum install java -y

```

  ____      _
 / ___|    / \   _ __| |__| |
| |  | |  / _ \ / _ \| |__| | | |
| |  | | / ___ \| |__| |__| |
| |  | |/_/   \_\_\_|_|__|_|
|_|  |_|

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Dec 2 14:54:28 2025 from 13.48.4.202
[ec2-user@ip-172-168-23-94 ~]$ sudo -i
[root@ip-172-168-23-94 ~]# yum install java -y
Last metadata expiration check: 0:00:37 ago on Tue Dec 2 14:58:09 2025.
Dependencies resolved.

```

Package	Architecture	Version	Repository
Installing:			
java-25-amazon-corretto	x86_64	1:25.0.1+8-1.amzn2023.1	amazonlinux
Installing dependencies:			
alsa-lib	x86_64	1.2.7.2-1.amzn2023.0.2	amazonlinux
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux
fontconfig	x86_64	2.13.94-2.amzn2023.0.2	amazonlinux
fonts-filesystem	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux
freetype	x86_64	2.13.2-5.amzn2023.0.1	amazonlinux
giflib	x86_64	5.2.1-9.amzn2023.0.2	amazonlinux
google-noto-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux

curl -O <https://dldcn.apache.org/tomcat/tomcat-9/v9.0.112/bin/apache-tomcat-9.0.112.tar.gz>

```
[root@ip-172-168-23-94 ~]# curl -O https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.112/bin/apache-tomcat-9.0.112.tar.gz
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 12.4M  100 12.4M    0     0  132M    0  --:--:--  --:--:--  --:--:-- 133M
[root@ip-172-168-23-94 ~]# ls
apache-tomcat-9.0.112.tar.gz
[root@ip-172-168-23-94 ~]#
```

tar -xzf apache-tomcat-9.0.112.tar.gz -C /opt

```
[root@ip-172-168-23-94 ~]# tar -xzf apache-tomcat-9.0.112.tar.gz -C /opt
apache-tomcat-9.0.112/conf/
apache-tomcat-9.0.112/conf/catalina.policy
apache-tomcat-9.0.112/conf/catalina.properties
apache-tomcat-9.0.112/conf/context.xml
apache-tomcat-9.0.112/conf/jaspic-providers.xml
apache-tomcat-9.0.112/conf/jaspic-providers.xsd
apache-tomcat-9.0.112/conf/logging.properties
apache-tomcat-9.0.112/conf/server.xml
apache-tomcat-9.0.112/conf/tomcat-users.xml
apache-tomcat-9.0.112/conf/tomcat-users.xsd
apache-tomcat-9.0.112/conf/web.xml
apache-tomcat-9.0.112/bin/
apache-tomcat-9.0.112/lib/
apache-tomcat-9.0.112/logs/
apache-tomcat-9.0.112/temp/
apache-tomcat-9.0.112/webapps/
apache-tomcat-9.0.112/webapps/ROOT/
apache-tomcat-9.0.112/webapps/ROOT/WEB-INF/
```

cd /opt/apache-tomcat-9.0.112/webapps

```
[root@ip-172-168-23-94 ~]# cd /opt
[root@ip-172-168-23-94 opt]# ls
apache-tomcat-9.0.112 aws
[root@ip-172-168-23-94 opt]# cd apache-tomcat-9.0.112/
[root@ip-172-168-23-94 apache-tomcat-9.0.112]# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[root@ip-172-168-23-94 apache-tomcat-9.0.112]# cd webapps/
[root@ip-172-168-23-94 webapps]# ls
ROOT docs examples host-manager manager
[root@ip-172-168-23-94 webapps]# pwd
/opt/apache-tomcat-9.0.112/webapps
[root@ip-172-168-23-94 webapps]#
```

curl -O <https://s3-us-west-2.amazonaws.com/studentapi-cit/student.war> This Link is Provide by Developer. This is My Frontend

```
[root@ip-172-168-23-94 webapps]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/student.war
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 89423  100 89423    0     0  104k    0  --:--:--  --:--:--  --:--:-- 104k
[root@ip-172-168-23-94 webapps]#
```

cd ..

cd lib

curl -O <https://s3-us-west-2.amazonaws.com/studentapi-cit/mysql-connector.jar> This Link is Provide by Developer. This is My Backend

```
[root@ip-172-168-23-94 ~]# cd /opt/apache-tomcat-9.0.112/webapps
[root@ip-172-168-25-106 webapps]# cd ..
[root@ip-172-168-25-106 apache-tomcat-9.0.112]# cd lib
[root@ip-172-168-25-106 lib]# ls
annotations-api.jar catalina-tribes.jar jasper.jar tomcat-coyote-ffm.jar tomcat-i18n-es.jar tomcat-i18n-ru.jar tomcat-util.jar
catalina-ant.jar catalina.jar jaspic-api.jar tomcat-coyote.jar tomcat-i18n-fr.jar tomcat-i18n-zh-CN.jar tomcat-websocket.jar
catalina-ha.jar ecj-4.20.jar jsp-api.jar tomcat-dbcp.jar tomcat-i18n-jp.jar tomcat-jdbc.jar tomcat-websocket-api.jar
catalina-ssi.jar el-api.jar servlet-api.jar tomcat-i18n-cs.jar tomcat-i18n-ko.jar tomcat-jni.jar
catalina-storeconfig.jar jasper-el.jar tomcat-api.jar tomcat-i18n-de.jar tomcat-i18n-pt-BR.jar tomcat-util-ssan.jar
[root@ip-172-168-25-106 lib]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/mysql-connector.jar
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 983k  100 983k    0     0  672k    0  0:00:01  0:00:01  --:--:-- 672k
[root@ip-172-168-25-106 lib]# ls
annotations-api.jar catalina-tribes.jar jasper.jar tomcat-api.jar tomcat-i18n-de.jar tomcat-i18n-pt-BR.jar tomcat-util-ssan.jar
catalina-ant.jar catalina.jar jaspic-api.jar tomcat-coyote-ffm.jar tomcat-i18n-es.jar tomcat-i18n-ru.jar tomcat-util.jar
catalina-ha.jar ecj-4.20.jar jsp-api.jar tomcat-coyote.jar tomcat-i18n-fr.jar tomcat-i18n-zh-CN.jar tomcat-websocket.jar
catalina-ssi.jar el-api.jar mysql-connector.jar tomcat-dbcp.jar tomcat-i18n-jp.jar tomcat-jdbc.jar websocket-api.jar
catalina-storeconfig.jar jasper-el.jar servlet-api.jar tomcat-i18n-cs.jar tomcat-i18n-ko.jar tomcat-jni.jar
[root@ip-172-168-25-106 lib]#
```

cd ..

ls

cd conf/

ls

```
[root@ip-172-168-23-94 lib]# cd ..
[root@ip-172-168-23-94 apache-tomcat-9.0.112]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-168-23-94 apache-tomcat-9.0.112]# cd /conf
-bash: cd: /conf: No such file or directory
[root@ip-172-168-23-94 apache-tomcat-9.0.112]# cd conf/
[root@ip-172-168-23-94 conf]# ls
catalina.policy      context.xml          jaspic-providers.xsd  server.xml            tomcat-users.xsd
catalina.properties  jaspic-providers.xml  logging.properties   tomcat-users.xml      web.xml
[root@ip-172-168-23-94 conf]#
```

vim context.xml

add the following content in this file

```
<Resource name="jdbc/TestDB" auth="Container" type="javax.sql.DataSource"
    maxTotal="500" maxIdle="30" maxWaitMillis="1000"
    username="admin" password="sakshi123" driverClassName="com.mysql.jdbc.Driver"
    url="jdbc:mysql://RDS      database      endpoint:3306/mysql      database
name?useUnicode=yes&characterEncoding=utf8"/>
```

```
[root@ip-172-168-25-106 conf]# vim context.xml
```



15. Start Tomcat

```
cd /opt/apache-tomcat-9.0.112/bin
./startup.sh
```

```
[root@ip-172-168-25-106 ~]# cd /opt/apache-tomcat-9.0.112
[root@ip-172-168-25-106 apache-tomcat-9.0.112]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-168-25-106 apache-tomcat-9.0.112]# cd bin
[root@ip-172-168-25-106 bin]# ^C
[root@ip-172-168-25-106 bin]# ls
bootstrap.jar      catalina.sh      commons-daemon-native.tar.gz  configtest.sh  digest.sh      setclasspath.bat  shutdown.sh  tomcat-juli.jar      tool-wrapper.sh
catalina-tasks.xml  ciphers.bat      commons-daemon.jar           daemon.sh      makebase.bat   setclasspath.sh   startup.bat   tomcat-native.tar.gz  version.bat
catalina.bat       ciphers.sh       configtest.bat              digest.bat     makebase.sh    shutdown.bat      startup.sh   tool-wrapper.bat     version.sh
[root@ip-172-168-25-106 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.112
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.112
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.112/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.112/bin/bootstrap.jar:/opt/apache-tomcat-9.0.112/bin/tomcat-juli.jar
Tomcat started.
[root@ip-172-168-25-106 bin]#
```

16. From your laptop browser:

http://<App_Server_Public_IP>:8080/student

Student Registration Form

Student Name

Student Address

Student Age

Student Qualification

Student Percentage

Year Passed

Student Registration Form

Student Name

Student Address

Student Age

Student Qualification

Student Percentage

Year Passed

[Register Student](#)

Students List

Student ID	StudentName	Student Addr	Student Age	Student Qualification	Student Percentage	Student Year Passed	Edit	Delete
1	Test Name	Test Addr	22	BE	75	2024	edit	delete
2	Gunjan	Umarane	22	Engg	85	2025	edit	delete
3	Sakshi Jain	Chandwad	22	BE	87	2025	edit	delete

```
mysql -h <RDS-endpoint> -u mydblogin -p
```

```
USE studentapp;
```

```
SELECT * FROM students;
```

ou should see all records you added from the application → **proves 2-tier flow is working:**

- Browser → Tomcat → RDS.

```
[root@ip-172-168-25-106 ~]# mysql -h database-1.c9wsoui0ei4j.eu-north-1.rds.amazonaws.com -u admin -psakshi123
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 219
Server version: 11.4.8-MariaDB-log managed by https://aws.amazon.com/rds/
```

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
MariaDB [(none)]> use studentapp;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

Database changed

```
MariaDB [studentapp]> show tables;
```

```
+-----+
| Tables in studentapp |
+-----+
| student              |
| students              |
+-----+
2 rows in set (0.001 sec)
```

```
MariaDB [studentapp]> select * from students;
```

```
+-----+-----+-----+-----+-----+-----+-----+
| student_id | student_name | student_addr | student_age | student_qual | student_percent | student_year_passed |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Test Name | Test Addr | 22 | BE | 75 | 2024 |
| 2 | Gunjan | Umarane | 22 | Engg | 85 | 2025 |
| 3 | Sakshi Jain | Chandwad | 22 | BE | 87 | 2025 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```